Nursing Informatics for Quality Improvement

By

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Definition of Health Data: “Health data refers to data related to a person’s health; their health conditions, information relating to maternity and children, causes of death, and quality of life. Health data includes, for example: patient health records, studies about the health of groups of people, data from blood or tissue samples.

Health informatics: is a multidisciplinary field that includes computer science, information science, and healthcare. It focuses on the use of information technology to improve the quality of healthcare.

Nursing informatics:

- Nursing informatics (Information Technology (IT) in Nursing) refers to the application of computer systems and software to improve patient care, enhance effectiveness, sufficiency, and make informed decisions in the field of nursing.
- It involves the use of technology to manage and process health information, communicate with patients and healthcare providers, and support clinical decision-making

Advantages of Nursing Informatics:

- Support for their mission to deliver high quality, evidence-based care.

- Privacy, confidentiality, and security of health information.

- Promote improvement in key relationships with physicians, peers, patients, families, and interdisciplinary team members.

- Decreased time spent on documentation.

- Systematically collect data that can be transformed into information for decision making.

- Save Time and effort spent in Paper storage printouts.

Disadvantages of Nursing Informatics: -

Computer malfunction.

Impersonal effect: by making information available to anyone.

Concern for privacy: information can conceivably be accessed or destroyed by unauthorized people.

Dissemination of inaccurate information.

Limitation of standard vocabulary: Standardized and limited vocabulary or phrase structure in computer formats can be a restriction on accuracy

It will be many years until health care organization can move to paperless chart. Save time for patient care.

Cost: cost of the hardware and software, training, supervisory

Role of informatics nurse

1. Improve nursing practice and the quality of patient care.

2. Facilitation of data collection for research.

3. Support decision based on assessments to drive a new plane of care.

4. Support for cost saving and productivity goals.

5. Creating new tools for patient care from new technologies

6. Integrating systems

7. Evaluating the effects of nursing systems.

Barriers to nursing informatics:

- Limited financial resources, as no budget is enough to by computers and other machines.

- Resistance to change as many years' nurses use paper in documentation very difficult for them when they are old to use electronic devices

- Needs training and education to the nurses to know how to use new system or devices.

- Fear of computerization, most nurses are afraid to make mistakes while using advanced machines or computers.

- Time management, nurses need more time to accommodate new devices, and they will face an issue regarding time management.

Types of Nursing information system

Electronic Medical Record (EMR):

An electronic version of traditional record used by health care providers. It includes information about a patient’s health history, such as diagnoses, medicines, tests, allergies, immunizations, and treatment plans.

Electronic Health Records (EHRs):

Digital versions of paper charts
Store patient medical history, medications, allergies, immunization records, and more.

- **Which can**
- It can be accessed from several different locations simultaneously, as well as by different levels of providers.
- Decision support tools as well as alerts and reminders notify the clinician of possible concerns or omissions. An example of this is the documentation of the patient allergies in the computer system. Health care providers would be alerted to any discrepancies in patient medication orders.
- Facilitate data collection for research: electronically stored client records provide quick access to clinical data for many clients.

Clinical Decision Support Systems (CDSS):

- Computer programs that analyze patient data and provide evidence-based recommendations.
- Help nurses make informed decisions about diagnosis and treatment plans.
- Reduce medical errors and improve patient outcomes.

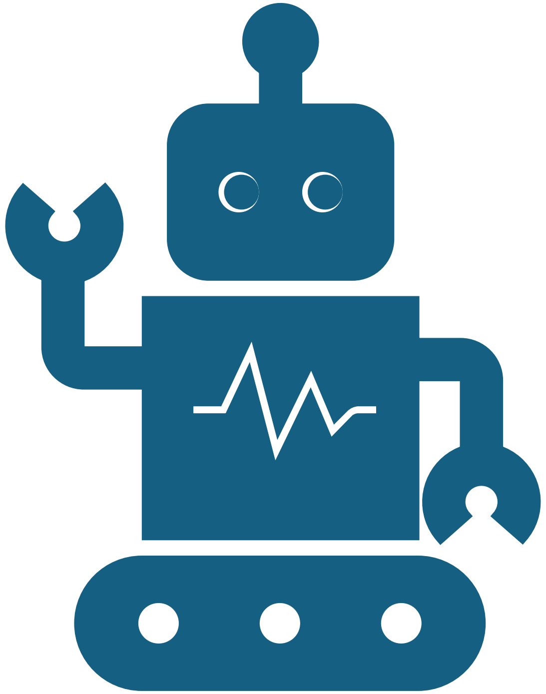
- **Telehealth and Telemedicine:**
- Use technology to deliver healthcare services remotely
- Enable virtual consultations, remote monitoring, and telemedicine appointments
- Expand access to care, especially for rural and underserved populations
- **Coaching**: through using AI in training and educating nursing staff.
- **Wearable Technology:**
- Devices that track various health metrics, such as heart rate, blood pressure, and activity levels
- Provide real-time data to healthcare providers and patients
- Promote preventive care and self-management .

Big Data Analytics

Analyze large datasets to identify patterns and trends Predict potential health risks and optimize resource allocation.

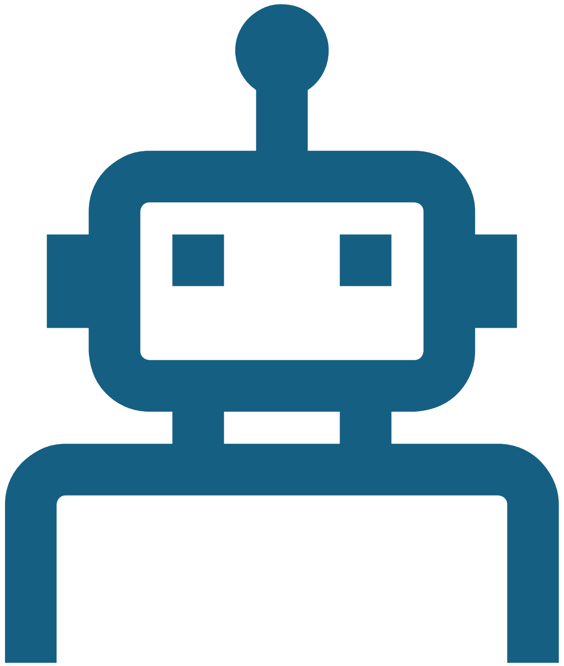
Improve population health management.

Robots in health care system (AMR, Modular Robots and Surgical-Assistance Robots)



- Robots in the medical field are playing an important role. Intel offers a diverse portfolio of technology for the development of medical robots, including surgical-assistance, modular, and autonomous mobile robots, which streamline supply delivery and disinfection, transforming how surgeries are performed and enabling providers to focus on engaging with and caring for patients.

Types of Robotics in Healthcare



- Using **robotics** in the medical field enables a high level of patient care, efficient processes in clinical settings, and a safe environment for patients and healthcare workers.
- **Autonomous mobile robots (AMRs).**
- **High-Quality Patient Care .**
  - Medical robots support minimally invasive procedures, customized and frequent monitoring for patients with chronic diseases.
- In addition, robots alleviate nursing & caregivers workloads

Streamlined Clinical Workflows

- Simplify routine tasks, reduce the physical demands on human workers, and ensure more consistent processes.
- These robots can address staffing shortages.
- help make sure supplies, equipment, and medication are in stock where they are needed.
- Cleaning and disinfection AMRs enable hospital rooms to be sanitized and ready for incoming patients quickly, allowing workers to focus on patient-centric, value-driven work.

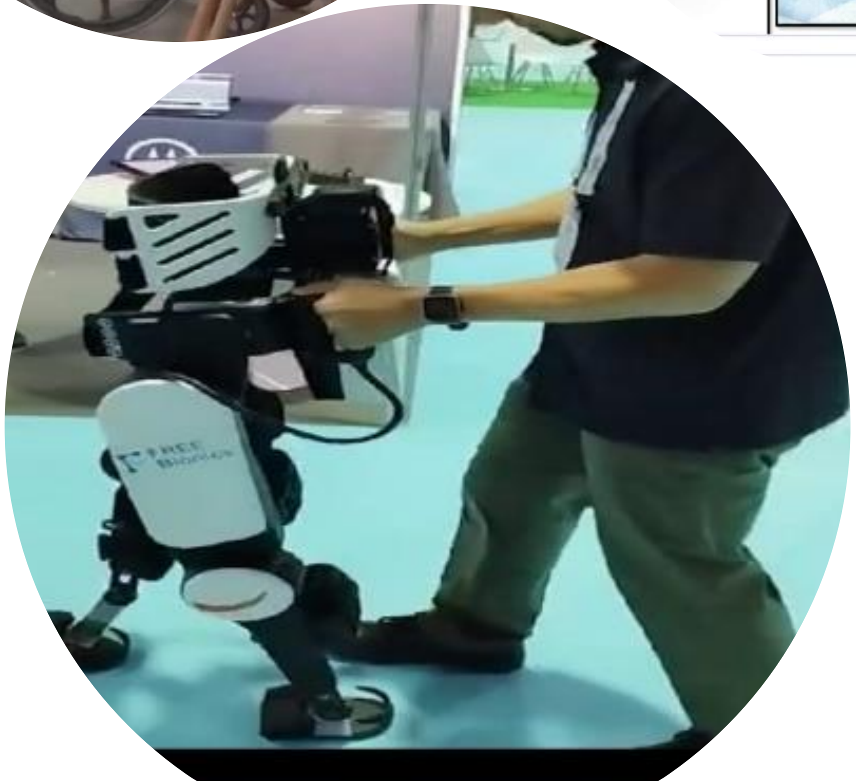
- **Safe Work Environment**
  - To help keep healthcare workers safe, AMRs are used to transport supplies and linens in hospitals where pathogen exposure is a risk.
  - Cleaning and disinfection robots limit pathogen exposure while helping reduce hospital acquired infections (HAIs)—and hundreds of healthcare facilities are already using them.
  - Social robots, a type of AMR, also help with heavy lifting, such as moving beds or patients, which reduces physical strain on healthcare workers.

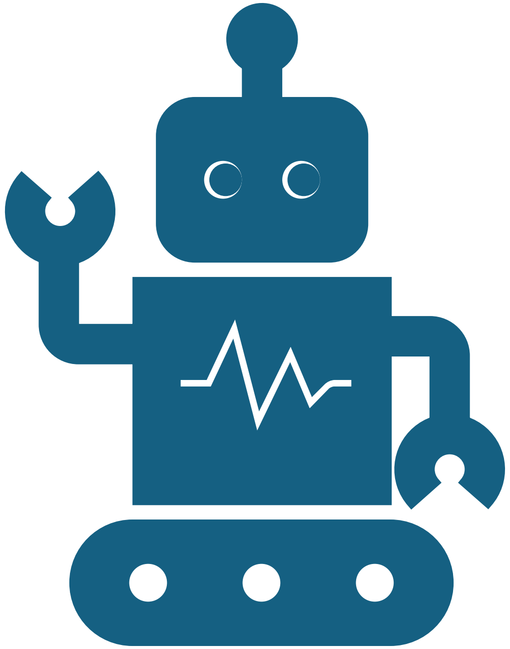
Surgical-Assistance Robots

- 1.These robots help surgeons achieve new levels of speed and accuracy while performing complex operations with AI- and computer vision–capable technologies.
- 2. Some surgical robots may even be able to complete tasks autonomously, allowing surgeons to oversee procedures from a console.
- 3. Simulation platforms use AI and virtual reality to provide surgical robotics training. Within the virtual environment, surgeons can practice procedures and hone skills using robotics controls.

Modular Robots

- Modular robots enhance other systems and can be configured to perform multiple functions.
- 1. In healthcare, these include therapeutic exoskeleton robots and prosthetic robotic arms and legs.
- Therapeutic robots can help with rehabilitation after strokes, paralysis, or traumatic brain injuries or with impairments caused by multiple sclerosis.
- A wheelchair-mounted robotic arm currently being developed by Intel aims to assist patients with spinal injuries in performing daily tasks.





Applying Autonomous Mobile Robots to Real-World

- . Some robots can assist professionals before a patient is checked in. For example, one autonomous robot in Mexico, developed by start-up Roomie, is being used to help medical staff with high-risk COVID-19 patients. Named Roomie Bot, it triages patients by taking their temperature, blood oxygen level, and medical history when they arrive at the hospital. Roomie Bot uses Intel-based technology.
- The Future of Robotics and Healthcare
- Health robotics will continue to evolve alongside advancements in machine learning, data analytics, computer vision, and other technologies. Robots of all types will continue to evolve to complete tasks autonomously, efficiently, and accurately.